

Airbnb NYC Data Analysis Report

Pricing Trends, Room Type Preferences & Neighbourhood Insights (2024)

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Objective

The objective of this project is to analyse Airbnb listings in New York City to understand pricing trends, room type preferences, listing availability, and neighbourhood-wise distribution using Python, SQL, and Power BI. The insights are intended to help hosts and business stakeholders make more informed pricing and marketing decisions.

Dataset Overview

Attribute	Details
Dataset Name	AB_NYC_2024.csv
Total Listings	20,758
Tools Used	Python, Pandas, SQL (SQLite), Power BI, GitHub

Dashboard Overview

The Power BI dashboard consolidates the analysis into the following components:

- Total Listings
- Average Price
- Average Rating
- Average Availability
- Listings by Room Type
- Room Type Distribution
- Average Price by Borough
- Average Availability by Borough
- Interactive Filters

Key Insights

1. Room Type Distribution

- Entire Home/Apartment is the most popular room type.
- Private Rooms are the second most common option.
- Shared Rooms and Hotel Rooms represent a very small percentage of listings.

2. Pricing Analysis

- Manhattan has the highest average Airbnb prices.
- Brooklyn follows with moderately high prices.
- The Bronx and Staten Island offer comparatively affordable listings.

3. Availability Analysis

- Availability differs noticeably across boroughs.

- Some boroughs show higher average availability, indicating lower occupancy or seasonal demand.

4. Customer Ratings

The average rating across listings is 4.73, indicating generally positive guest experiences.

20,758 Total Listings	4.73 Avg. Rating	Manhattan Top Borough	Entire Home/Apt Most Popular Type
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Recommendations

- Hosts in Manhattan can maintain premium pricing due to consistently higher demand.
- Hosts in lower-priced boroughs should focus on improving amenities and customer experience to attract more guests.
- Regularly monitoring availability data can help hosts optimize pricing strategy and occupancy rates.

Conclusion

This project demonstrates how data analytics can be used to identify market trends, understand customer preferences, and support data-driven decision-making in the Airbnb market.

The combined use of Python, SQL, and Power BI enabled effective data cleaning, exploration, visualization, and reporting throughout the analysis.